

- Q.30 Draw neat and labeled sketch of blast furnace.
- Q.31 Explain bio-chemical conversion of biomass in brief.
- Q.32 Explain working of vertical axis wind turbine with advantages and disadvantages.
- Q.33 Explain Tidal energy with their merits and demerits in details.
- Q.34 Describe the working of solar collector.
- Q.35 Explain in brief the Geo-thermal energy with its merits and demerits.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe the high temperature carburization process.
- Q.37 With neat sketches, explain the working principle of continuous rotary kiln used in the manufacturing of Portland cement.
- Q.38 With neat sketch and labeled block diagram, explain the working of hydraulic power plant and state its merits and demerits over thermal power plants.

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Roll No.

4th Sem / Chem Subject:- Energy Technology

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which are the main constituents of Fuels?
a) Carbon & Nitrogen b) Oxygen & Hydrogen
c) Carbon & Hydrogen d) Helium & Hydrogen
- Q.2 What is the significance of calorific value?
a) Help in deciding which fuels are good
b) Helps in locating flues
c) Helps in deciding ignition temperature
d) Helps in deciding fire point
- Q.3 Fischer - Tropsch method aim's of
a) Gasification of coal
b) Synthesis of gasoline
c) Hydrogenation of coal to produce gasoline
d) None
- Q.4 Crude oil is
a) Colorless
b) Odorless
c) Smelly yellow to black liquid
d) Odorless yellow to black liquid

- Q.5 Producer gas is mixture of
 a) Carbon-mono-oxide & Nitrogen
 b) Carbon-mono-oxide & Hydrogen
 c) Carbon-mono-oxide & oxygen
 d) Hydrogen & Nitrogen
- Q.6 At normal temperature and pressure, in which form L.P.G is found?
 a) Solid b) Gaseous
 c) Liquid d) Solid-liquid
- Q.7 What is the boiling temp. of petrol?
 a) 40-120° C b) 120-180° C
 c) 180-250° C d) 250-320° C
- Q.8 What is the percentage of methane in Bio-gas?
 a) 10-20% b) 20-40%
 c) 45-65% d) 55-75%
- Q.9 Which of the following is an advantage of most of renewable energy sources?
 a) Less polluting
 b) High waste disposal cost
 c) Unreliable supply
 d) High running cost
- Q.10 Which fuel is widely used in steam power plants?
 a) Oil b) Gas
 c) Coal d) Petroleum

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Name two conventional fuels.

- Q.12 Name two methods of coal mixing.
- Q.13 Write types of coal in increasing order of their calorific value.
- Q.14 Write the names of theories of petroleum origin.
- Q.15 Write two functions of furnaces.
- Q.16 Write the calorific value of L.P.G.
- Q.17 Name two products for which continuous rotary kiln used.
- Q.18 What is photosynthesis.
- Q.19 Expand HWT and VWT.
- Q.20 Name two thermal power plants in India.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Differentiate between conventional and non-conventional energy sources.
- Q.22 Define calorific value.
- Q.23 Differentiate between low and high temperature carbonization.
- Q.24 Explain principle and objective of coal washing.
- Q.25 Explain distillation of crude oil and name various products obtained.
- Q.26 Explain carbide theory in details.
- Q.27 Describe manufacturing of producer gas and state its uses.
- Q.28 Describe manufacturing of water gas.
- Q.29 Explain the working principle of glass melting furnace.

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